MC PROJECT · INNOVATION PIPELINE · PHASE 2

Patent Landscape Analysis

Prior-art, white-space and portfolio assessment of five embedded-systems concepts:
OpenBraille · VibeGuard · TrueMoist · TrustLatch · ColdTrace

Prepared for the Engineering R&D Decision Board

Evidence base: 121 indexed sources · USPTO full-text records · peer-reviewed and program literature

As of 17 July 2026 · Final for board review · Not legal advice / not an FTO opinion

Contents

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Scope note and legal disclaimer. This report is a technology-landscape and white-space assessment prepared to support R&D prioritization. It is not a freedom-to-operate (FTO) opinion, not a validity study, and not legal advice. Patent identifiers, titles, and priorities were verified against full-text records; expiry estimates are indicative only and must be confirmed by counsel (term adjustments, continuations, and family members can shift dates). Several primary databases (Google Patents, Espacenet, USPTO Patent Public Search, InPASS) were partially access-restricted during this pass; US records were therefore retrieved through a mirrored full-text index, and JP/CN/IN coverage is representative rather than exhaustive. Where evidence is thin, the gap is stated explicitly in each concept's Section 12.

Executive Summary

Phase 1C fixed five concepts for landscape analysis. This phase subjected each to a twelve-section assessment covering problem framing, commercial ecosystem, patent and prior-art density, technical feasibility, white space, defensibility, regulatory pull, risk, and confidence. Three structural findings cut across the portfolio.

Finding 1 — The strongest opportunities sit where foundational patents have just expired, not where the field is empty. OpenBraille is the clearest case: the shape-memory-alloy cell (US 5,685,721, 1997) [27] and the electroactive-polymer cell (US 6,881,063, 2005) [26] have both lapsed into the public domain, and the Orbit Reader 20 has already proven that a non-piezoelectric cell can ship at **$449** against a $3,500–$15,000 incumbent range [2][107]. ColdTrace shows the same pattern in algorithms: the electronic time-temperature "stability bank" indicator (US 7,102,526, 2003 priority) [116] is at/past expiry, freeing model-based excursion logic — even as one active patent (US 10,887,735) fences off the buffered-probe implementation [117].

Finding 2 — Two concepts face red-ocean patent density where further invention is low-yield. VibeGuard's bearing-fault-detection space is saturated: signature-detection methods explicitly robust to unrelated spectral peaks were already patented by 2008 (US 7,421,349) [118], platform-scale IIoT analytics families and a dense stream of CNN-based filings crowd the detection layer [46][47], and a venture-backed incumbent (Infinite Uptime) already occupies the SME wedge with patented technology [54][56]. TrustLatch's *core cryptography* is equally unpatentable

— but deliberately so, because MCUboot/TF-M/Mender have made it open infrastructure [75][79][80]. Its opportunity is compliance-driven usability, not IP.

Finding 3 — Regulatory calendars now differentiate the portfolio more than technology does. TrustLatch rides the EU Cyber Resilience Act / UK PSTI wave that is converting secure update from option to obligation [74][76]; ColdTrace rides WHO/CDC vaccine-handling rules and GDP [93][96][100]; OpenBraille rides accessibility funding channels (APH Quota, ADIP) [109][110]. VibeGuard and TrueMoist have no equivalent forcing function, which lengthens their sales cycles.

Composite ranking (equal-weighted six-dimension score, 0–10):

Rank	Concept	Composite	Patent saturation	White space	Headline recommendation
1	OpenBraille	**7.0**	Moderate	Moderate–High	Advance to Phase 3; file provisional on low-cost actuation kinematics
2	TrustLatch	**6.8**	Moderate (mis-aimed at PCs/servers)	Moderate–High (usability, not crypto)	Advance as open-core compliance toolkit; speed over patents
3	ColdTrace	**6.3**	Moderate–High	Moderate (design-around required)	Advance with FTO review of US 10,887,735; virtual thermal mass
4	TrueMoist	**5.5**	Moderate	Low–Moderate	Hold; validate calibration-free field performance first
5	VibeGuard	**5.0**	High	Low	Demote; revisit only with a sharper vertical wedge

Scores are analyst assessments on a 0–10 ordinal scale derived from the evidence assembled in Sections 1–5 of this report; they are prioritization aids, not measurements. The full scoring rubric and cross-concept evidence matrix appear in Part 6.

Method & Evidence Conventions

Each concept chapter follows the mandated twelve-section template: (1) Problem Summary, (2) Commercial Ecosystem Scan, (3) Patent & Prior-Art Landscape, (4) Technical Prior-Art and Feasibility, (5) White-Space Assessment, (6) Competitive & Defensibility Analysis, (7) Regulatory / Standards Context, (8) Risk Matrix, (9) Recommended Next Step, (10) Key Patent & Prior-Art Table, (11) Commercial Landscape Table, (12) Confidence Score & Evidence Gaps. Patent records were read at full-text level for the highest-relevance documents in each domain — nine patents in depth (US 5,685,721; US 6,881,063; US 11,410,574; US 2024/0363022; US 7,884,620; US 11,598,743; US 7,102,526; US 10,887,735; US 7,421,349; US 10,740,084; US 9,953,166; US 11,042,609), with hundreds of further records screened through result-set metadata [25][46][47][68][69][82][83][88][89].

Three conventions apply throughout. Citations use bracketed source IDs that resolve to the reference annex; a maximum of two IDs is attached to any single claim. Patent expiry language is deliberately hedged ("expected", "at/near expiry") because exact term depends on adjustments and family data outside this pass. And every confidence score in Section 12 of each chapter is accompanied by the evidence gap that most threatens it, so the board can see what an additional research sprint would need to close.

1. OpenBraille — Low-Cost Refreshable Braille Display Module

1.1 Problem Summary

Refreshable braille displays (RBDs) are the only channel through which blind users can access text — and, via multiline variants, spatial content such as tables, math, and graphics — in a non-transient, literacy-preserving form. The category has been structurally expensive for four decades: the American Foundation for the Blind places typical single-line display prices at **$3,500 to $15,000** [2], a price band widely corroborated by user communities and charities that frame affordability as the field's central barrier [8][9]. The cost driver is the actuation cell: incumbent devices use piezoelectric bimorph stacks whose per-cell cost has historically been estimated in the $100-plus range, so a 40-cell line embeds thousands of dollars of actuators before electronics, housing, or margin [11].

The user need splits into two segments with different willingness-to-pay structures. In wealthy markets, users and agencies want cheaper single-line companion displays and, increasingly, multiline tactile-graphics capability [14][22]. In low- and middle-income markets — where the majority of the world's blind population lives and where braille literacy correlates strongly with employment — the need is an order-of-magnitude price break with offline content loading, exactly the profile that led a Transforming Braille Group consortium including RNIB to back the Orbit Reader program [109]. OpenBraille, as framed in Phase 1C, targets a **low-cost refreshable braille display module** — the cell-plus-driver building block — rather than a finished consumer device, positioning it to serve OEMs, education programs, and the multiline frontier simultaneously.

1.2 Commercial Ecosystem Scan

The commercial landscape has bifurcated since ~2017 into an incumbent piezoelectric tier and a disruptor tier built on alternative actuation. In the disruptor tier, the Orbit Reader 20 sells at **$449** (20 cells) and its Plus variant at **$899** [107][112]; Innovision's Braille Me, developed in India and distributed by National Braille Press, sells at **$515.50** and is marketed as built on patented cell technology [110][34]. Orbit Research describes its cell as using "mainstream components and manufacturing processes" to deliver signage-quality dots "at a fraction of the cost of the piezoelectric technology" [113]. Bristol Braille's Canute line pursues multiline reading through an open-source, community-supported model [36][37][39], and the Blitab project demonstrated sustained investor/press appetite for a ~$500 tactile tablet concept [1][5].

At the premium end, HumanWare's Monarch multiline tablet — built on Dot Incorporation's actuator cells in partnership with APH — lists at **$15,500** [15][20], while Dot's own Dot Pad X targets tactile graphics for developers and prosumers [19][16]. The surrounding software ecosystem (JAWS at $895, NVDA free, VoiceOver/BrailleBack) is display-agnostic, so hardware differentiation rests on cell cost, dot quality, refresh speed, and reliability [107]. The ecosystem read: **the sub-$1,000 single-line segment is proven but thinly supplied**, and the multiline segment is at its expensive infancy — both are module-opportunity windows rather than finished-device windows (see Figure 3).

1.3 Patent & Prior-Art Landscape

The patent record shows a 50-year cadence of actuation approaches, with the oldest now expiring into a usable public-domain base. Representative anchors read at full text: **US 5,685,721** (granted 1997) — a refreshable braille cell implemented with **shape-memory alloys**, now expired [27]; **US 6,881,063** (granted 2005) — an **electroactive-polymer** actuator braille cell, expected expired (~2022) [26]; **US 11,410,574** (granted 2022) — a **layered electromagnetic** braille display structure, active [29]; and **US 2024/0363022 A1** — a refreshable braille display system published in 2024, pending [28]. The surrounding result set is dense with expired or aging mechanism patents — rotary/wheel displays (e.g., US 7,744,372; US 8,133,055; US 6,776,619), reed-based cells (US 9,424,759; US 8,690,576), an early system patent US 6,354,839 (2002), and electrothermal cell concepts — alongside a continuing trickle of new filings, including a 2026 publication on braille display cell systems, indicating the field is technically re-activating after the multiline wave [25].

Two portfolio-scale players matter most for freedom-to-operate. **Dot Incorporation** has built a substantial patent estate around its miniaturized electromagnetic latching actuator ("Dot cell") and supplies the cells inside the Monarch; any multiline or latching-pin approach must be claim-charted against this family [19][17]. **Orbit Research** protects its low-cost cell architecture — the mechanism behind the $449 display — meaning OpenBraille cannot simply clone the proven disruptor either [113]. The academic and maker prior-art layer is also thick: ultra-low-cost device designs appear in journals, ISEF projects, and design contests [3][4][6][33], and an ACM TOCHI survey of pin-array product architectures documents essentially every historical kinematic approach [21]. Net: the *concept* of a low-cost cell is old and heavily published; patentable residue lies in **specific new kinematics, latching schemes, drive electronics, or manufacturing methods** that clear both the expired art (which is free to use but also anticipates) and the two active estates.

1.4 Technical Prior-Art and Feasibility

Feasibility is the strongest in the portfolio because the market has already executed natural experiments at the target price point. The Orbit Reader 20 demonstrates that a non-piezo cell can meet everyday reading duty at $449 in a mainstream commercial product [107][111]; Braille Me demonstrates an India-developed alternative at $515.50 with routing buttons [110]; Canute demonstrates that a small team can ship multiline refreshable braille at all [36][38]. The known engineering constraints are equally well documented in user and reviewer communities: refresh rate, dot firmness ("signage quality"), acoustic noise, and cell durability are the axes on which cheap cells are judged against piezo [107][108].

Prior art defines the actuation design space almost completely: piezoelectric bimorph (incumbent, expensive, excellent tactility), electromagnetic latching (Dot; Orbit-variant EM), SMA (expired US 5,685,721 — slow, power-hungry, but cheap) [27], electroactive polymers (expired US 6,881,063 — materials risk) [26], electrothermal (US 2007/0020589 A1), pneumatic/microfluidic (US 12,555,492), and rotary/continuous-surface schemes [25]. A Phase 3 prototype can therefore start from public-domain mechanisms and concentrate invention on **cost-out engineering** — e.g., shared-driver multiplexing, stamped rather than machined parts, or a novel low-force latching geometry — rather than on discovering a new physics. Technical risk is moderate; the binding constraint is demonstrating piezo-comparable dot force and >10^7-cycle endurance at a cell cost under roughly $10–15, the threshold implied by the $449/20-cell market proof.

1.5 White-Space Assessment

White space is **moderate-to-high**, and it is structural rather than conceptual. The expired base (SMA, EAP, early rotary/system patents) creates a legal free zone for mechanism choice; the demand gap is verified by the fact that two products (Orbit, Braille Me) could not fully supply the sub-$1,000 segment they created [9][110]. The uncrowded zones with genuine novelty potential are: (a) **multiline-capable low-cost cells** — Dot's estate is the main fence here, and Monarch's $15,500 price shows the cost problem is unsolved [15]; (b) **module-level products** (cell + driver + reference firmware sold to OEMs/education programs) — no dominant merchant supplier exists; and (c) **manufacturing-method IP** (injection-molded cell arrays, automated pin assembly) where publication is thinner than mechanism patents [21][25].

Conversely, two zones are closed: single-line low-cost *devices* (Orbit/Braille Me own the reference designs and price expectations), and electromagnetic latching-pin multiline (Dot's active filings plus the layered-EM structure of US 11,410,574) [29][19]. The white-space verdict: pursue the module, not the gadget; pursue manufacturing innovation, not another wheel patent.

1.6 Competitive & Defensibility Analysis

Direct competitors in the module sense are few; device competitors are Orbit Research (cell IP + APH distribution), Innovision (patented low-cost cell), Dot (premium multiline cells), and the piezo incumbents (HumanWare, HIMS, BAUM, Help Tech) whose cost structure is locked to bimorph stacks [107][110][19][2]. Indirect competition is audio: free screen readers keep improving, and any RBD must justify its existence against "good-enough speech," which is why literacy-critical and education use-cases are the defensible beachhead [8][107].

Defensibility would rest on three layers: a **mechanism patent** on the specific low-cost kinematics (must clear US 11,410,574, Dot's estate, and the expired art that anticipates generic approaches); **process/manufacturing IP**, which competitors cannot easily design around once a $5–10 cell cost is demonstrated; and **ecosystem lock-in** through reference firmware and education-program relationships (NBP/APH-style channels) [34][109]. Speed matters: the 2024–2026 publication uptick (US 2024/0363022; a 2026 cell-system publication) shows others are re-entering the space [28][25].

1.7 Regulatory / Standards Context

Demand-side regulation is favorable and unusually concrete. US federal procurement (Section 508), the ADA, and the European Accessibility Act / EN 301 549 obligate accessible interfaces in public contexts, sustaining agency purchasing; APH Quota funds explicitly subsidize displays for US students — the Orbit Reader 20 is a Quota-eligible product, a material sales channel [110]. India's ADIP scheme performs the same function domestically, and RNIB's direct distribution role in the UK shows how charity-channel endorsement functions as a regulatory-adjacent demand lever [109].

On standards, braille cell geometry (dot height/diameter/spacing) follows established norms that constrain mechanical design but are public; there is no safety-certification moat (CE/FCC suffice for a module). The Marrakesh Treaty eases cross-border accessible-format content, indirectly expanding the content base that justifies display purchases. None of these factors block OpenBraille; several actively subsidize its target customers.

1.8 Risk Matrix

Risk	Likelihood	Impact	Mitigation

FTO conflict with Dot latching-actuator or layered-EM claims	Medium	High	Early claim-charting; choose kinematics outside latching EM; license if multiline pursued
Cannot hit piezo-comparable dot force/endurance at target cost	Medium	High	Phase 3 gate: 10^7-cycle + force test before any filing
Orbit/Braille Me extend downward, commoditizing the segment	Medium	Medium	Differentiate on module/OEM channel and multiline readiness, not on single-line devices
Screen-reader audio substitution shrinks addressable demand	Low–Med	Medium	Anchor in education/literacy and tactile-graphics use-cases [14][21]
Charity/agency procurement cycles slow revenue	High	Medium	Design for Quota/ADIP eligibility from day one [110][109]
Expired-art anticipation kills patent claims	Medium	Medium	File on manufacturing method + specific kinematics, not on "low-cost cell" as such [21][25]

1.9 Recommended Next Step

Advance to Phase 3. Build a cell-level prototype targeting <$15 cell cost at volume, using public-domain actuation physics, with invention concentrated on (i) a specific latching/drive kinematic cleared against US 11,410,574 and Dot's estate, and (ii) a manufacturability method (molded array, automated pin insertion). File a provisional on those two layers once the endurance gate (10^7 cycles, dot-force parity with piezo) is passed; in parallel, open OEM/education-channel conversations patterned on the NBP/Braille Me and APH/Orbit precedents [34][110].

1.10 Key Patent & Prior-Art Table

Document	Year	Holder / type	Content	Status & relevance
US 5,685,721	1997	SMA braille cell	Shape-memory-alloy refreshable cell [27]	Expired — free design base
US 6,881,063	2005	EAP braille cell	Electroactive-polymer actuator cell [26]	Expected expired — free design base
US 6,354,839	2002	Refreshable braille display system	Early system-level claims [25]	Expired/near — background
US 11,410,574	2022	Layered electromagnetic display	Stacked EM structure for pin actuation [29]	**Active — key FTO fence**
US 2024/0363022 A1	2024	Refreshable display system	Recent system filing [28]	Pending — monitor
Dot actuator estate	2015–	Dot Incorporation	Latching EM micro-actuator family behind Dot Pad/Monarch [19][17]	**Active — key FTO fence**
Rotary/wheel family (US 7,744,372; US 8,133,055; US 6,776,619)	2006–2011	Various	Continuous-surface refresh approaches [25]	Mostly expired — free, but anticipates

ACM TOCHI pin-array survey	2023	Academic	Architectural taxonomy of pin-array devices [21]	Non-patent prior art — anticipation risk
Ultra-low-cost device literature	2018–2020	Academic/maker	Servo/motor-based teaching devices [3][4][33]	Non-patent prior art

1.11 Commercial Landscape Table

Player	Product / role	Price point	Evidence
Orbit Research / APH	Orbit Reader 20 / 20 Plus	$449 / $899	[107][112]
Innovision / NBP	Braille Me (patented cell)	$515.50	[110][34]
HumanWare / APH / Dot	Monarch multiline tablet	$15,500	[15][20]
Dot Incorporation	Dot Pad X + cell supply	Premium	[19][16]
Bristol Braille	Canute multiline (open source)	Community-priced	[36][37][39]
Blitab	Tactile tablet project	~$500 target	[1][5]
Piezo incumbents (HumanWare, HIMS, BAUM, Help Tech)	Traditional 14–80-cell displays	$3,500–$15,000	[2]

1.12 Confidence Score & Evidence Gaps

Confidence: 8/10 — the highest in the portfolio. Price points, the expired-patent base, and the two active FTO fences are all verified at primary-source level; feasibility is market-proven.

Evidence gaps: (1) **Dot estate claim charts** — the exact claim scope of Dot's latching-actuator family needs professional charting before multiline work; (2) Orbit Research's own patent numbers were identified via product literature rather than direct record reads [113]; (3) Indian (InPASS) filings by Innovision and others were outside this pass's reach and may contain module-relevant art; (4) no cell-cost teardown of the Orbit/Braille Me mechanisms was available, so the <$15 cell-cost target is an inference from retail prices, not a measured bill of materials.

2. VibeGuard — Spatially-Isolated Bearing Fault Early-Warning Node

2.1 Problem Summary

Unplanned rotating-machinery downtime is a well-quantified economic drain, and vibration-based condition monitoring is the standard early-warning modality for bearing degradation. Predictive-maintenance (PdM) vendors promise double-digit reductions in downtime and maintenance cost [43][51], yet adoption concentrates in large plants: small and mid-sized enterprises face per-point costs, analytics complexity, and alarm fatigue that make classic PdM uneconomic — a gap explicitly targeted by newer entrants [54][55]. The problem VibeGuard addresses, as framed in Phase 1C, is sharper than "cheap PdM": a monitoring node bolted to one machine in a dense shop floor also **feels the vibration of neighboring machines** through floors, benches, and shafts, so generic low-cost nodes generate false alarms or miss early faults — a noise-discrimination problem that the bearing-fault literature has studied for decades [118][50].

The technical question is therefore not whether bearing faults are detectable (a solved problem at the research level) but whether a **single, cheap, self-contained node** can attribute spectral energy to its own machine with sufficient confidence to be trusted by a non-specialist operator. The severity of the cross-talk problem depends on mounting, machine spacing, and rotational-speed separation between neighbors — worst in dense SME workshops, which is exactly the commercial target.

2.2 Commercial Ecosystem Scan

The market stratifies into three tiers. At the top, **Augury** sells an enterprise machine-/process-health platform with proprietary sensors and has achieved top-tier industry recognition [43][44][45]; **Tractian** markets a similar sensor-plus-platform bundle with aggressive SMB-adjacent pricing [51]. Incumbent instrument houses (SKF, Fluke/PRÜFTECHNIK, Emerson/AMS) dominate wired and route-based monitoring, with per-point economics aimed at critical assets [50][46]. The third tier is the one VibeGuard would enter: **Infinite Uptime** (India) explicitly targets SME plants, publicizes patented diagnostic technology, and raised a **$35M** growth round on that positioning [54][56][52] — simultaneously validating the wedge and demonstrating that it is already defended.

Adjacent commoditization pressure comes from the component layer: industrial MEMS vibration sensors and MCU-class edge inference are now catalog parts, and research-grade open designs for wireless MEMS monitoring nodes appear regularly in the literature [42][48]. The ecosystem read: detection-as-a-feature is commoditizing fast; the scarce commodity is **trustworthy attribution** (whose fault is it?) in acoustically coupled environments — but the patent record below shows even that layer is heavily worked.

2.3 Patent & Prior-Art Landscape

This is the most saturated landscape of the five concepts. Two FreePatentsOnline result sets — "vibration monitoring bearing fault diagnosis" and "predictive maintenance vibration edge machine learning" — return dense layers of filings spanning decades, from the US Navy's **US 7,421,349** "Bearing fault signature detection" (2008) through automated-diagnosis-with-confidence publications (US 2016/0041070 A1), platform-scale industrial-data families (the US 2020/0150643 A1 cluster), and a continuing stream of CNN-based diagnosis filings into 2024–

2025 (e.g., US 12,067,486) [46][47]. The 2008 Navy patent is particularly instructive: it explicitly solves the problem of **distinguishing fault-generated spectral peaks from coincidental peaks produced by unrelated processes** — using phase coupling between sidebands — which is the exact discrimination mechanism VibeGuard's "spatial isolation" framing implies [118]. In other words, the core signal-processing claim is anticipated by 20-year-old art.

Ownership spans the US government (Navy), platform industrials (GE-family IIoT analytics), condition-monitoring specialists, and a heavy tail of Chinese university assignees on CNN/attention-based diagnosis [46][47]. Academic prior art compounds the saturation: envelope analysis/high-frequency resonance is textbook; noise-robust and cross-machine fault diagnosis is an active journal topic with hundreds of papers [42][48][49][50]. The practical consequence: almost any detection algorithm VibeGuard might ship is either anticipated, obvious over, or affirmatively claimed by existing art; patentable residue would have to live in an unusually specific mechanical or deployment innovation.

2.4 Technical Prior-Art and Feasibility

Feasibility of the node itself is high and unremarkable: triaxial MEMS accelerometers with ≥10 kHz bandwidth, MCU-class FFT/envelope computation, and BLE/LoRa/Wi-Fi backhaul are all catalog engineering, and reference designs abound [48][42]. The discriminating feature — call it **spatial attribution** — has three candidate mechanisms, each with established prior art: (i) **spectral separation** using known neighbor speeds/orders (classical order tracking); (ii) **phase-coupling tests** that verify whether sideband families originate from one physical process (patented, US 7,421,349, at/near expiry, hence free to use but not protectable) [118]; and (iii) **blind source separation / multi-sensor fusion**, which is journal-active but rarely reduced to a robust single-node product [48][49].

The honest engineering assessment: a *demonstration* of spatial attribution on a bench rig is achievable in weeks; a *robust product* that maintains attribution across variable-speed machines, belt-driven loads, and structurally coupled neighbors is a multi-year data problem — the same problem the entire industry already works on [50][51]. No physics barrier exists; no shortcut exists either. VibeGuard's concept is feasible but undifferentiated at the invention layer.

2.5 White-Space Assessment

White space is **low**. At the algorithm layer, signature detection, noise rejection, confidence scoring, and edge deployment are all heavily claimed [46][47][118]. At the business-model layer, the SME wedge that justified the concept is occupied by a funded incumbent marketing patented technology [54][56]. The residual gaps are narrow: (a) **ultra-low-cost single-node hardware** sold without a cloud subscription — a pricing stance more than an invention; (b) **installation-physics innovations** (mounts, mechanical isolation, coupling-aware placement guidance) that reduce cross-talk *mechanically* rather than algorithmically — comparatively under-published; and (c) vertical niches (low-speed machinery, specific industries) where generic platforms underperform [50].

None of these gaps supports a strong patent position; all support at best a lean commercial entry. For a portfolio whose currency is defensible novelty, VibeGuard offers the least.

2.6 Competitive & Defensibility Analysis

Competitors include platform vendors (Augury, Tractian), instrument incumbents (SKF, Fluke), the SME-focused Infinite Uptime, and — increasingly — free/cheap reference designs plus the customers' own maintenance intuition [43][51][54]. Defensibility through patents is weak: detection IP is crowded, and enforcement against algorithmic lookalikes is notoriously difficult. Defensibility through data is theoretically possible (labeled cross-machine datasets are scarce) but requires deployments to accumulate — a chicken-and-egg dynamic that funded competitors are already past [56].

The realistic moat, if any, is **deployment economics**: a node cheap enough to be a consumable, with zero-config commissioning. That is a supply-chain and UX achievement, not IP — and it is perishable the moment a platform vendor ships a $99 sensor.

2.7 Regulatory / Standards Context

No mandate drives adoption; PdM is economically self-justifying where it works. Relevant standards are ISO 10816/20816 (vibration severity evaluation — useful as an alarm-threshold anchor), ISO 13373 (condition-monitoring data communication), and IECEx/ATEX only if hazardous-area variants are pursued, which would multiply certification cost. The absence of a forcing function means long, ROI-proof-driven sales cycles in the SME segment — consistent with how Infinite Uptime sells (outcome-based, subscription) [54][55].

A secondary consideration: because the node processes operational data, enterprise customers increasingly impose security review on connected sensors — an argument for pairing any such product with a TrustLatch-class secure-update foundation, linking two concepts in this portfolio.

2.8 Risk Matrix

Risk	Likelihood	Impact	Mitigation
Detection claims anticipated by crowded prior art	High	High	Do not file on detection; limit IP to mechanical/deployment specifics [46][118]
Cross-machine false alarms erode trust in the node	Medium–High	High	Field-trial gate in a dense-machine environment before commitment
Funded incumbents compress the SME wedge	High	High	Compete on no-subscription economics or exit [56]
Platform vendors ship cheap hardware, killing the cost moat	Medium	High	Treat hardware margin as temporary; services/vertical focus
Long SME sales cycles with no regulatory pull	High	Medium	Outcome-based pricing; anchor on ISO 10816 alarms
Data-access catch-22 for attribution model training	Medium	Medium	Partnerships with shop-floor networks; public datasets for bootstrap [48]

2.9 Recommended Next Step

Demote. Do not fund a patent-track prototype. If strategic interest persists, re-scope toward one of the two defensible niches — mechanical cross-talk mitigation or a named vertical — and run a 90-day field validation of single-node attribution accuracy in a real dense-machine workshop before any further investment. Reuse the concept's security requirements as a TrustLatch pilot use-case instead.

2.10 Key Patent & Prior-Art Table

Document	Year	Holder / type	Content	Status & relevance
US 7,421,349	2008	US Navy	Phase-coupling fault-signature detection robust to unrelated peaks [118]	At/near expiry — free to use, anticipates the core claim
US 2016/0041070 A1	2016	Industrial assignee	Automatic fault diagnosis with confidence output [47]	Active family possible — FTO check
US 2020/0150643 A1 cluster	2020	Platform industrial (GE-family)	IIoT data collection/learning/streaming for asset monitoring [47]	Active — platform fence
US 12,067,486	2024/25	Recent filing	Fault-diagnosis method (CNN-class) [46]	Active — shows filing velocity
CNN/attention diagnosis filings	2019–2025	Chinese universities	Dense application layer [46]	Active — saturation signal
Noise-robust bearing diagnosis literature	2023	Academic (MDPI Sensors et al.)	Cross-condition/noise-robust diagnosis [48][42]	Non-patent prior art
Brunel condition-monitoring thesis	2015+	Academic	Field survey of techniques and vendors [50]	Non-patent prior art
Infinite Uptime patented IDE technology	—	Corporate (India SME)	Patented diagnostics, SME deployment model [54][52]	Competitor IP — FTO + market fence

2.11 Commercial Landscape Table

Player	Positioning	Business model	Evidence
Augury	Enterprise machine/process health	Sensor + SaaS subscription	[43][44][45]
Tractian	Sensor + platform, aggressive SMB pricing	Subscription	[51]
Infinite Uptime	SME-focused, patented diagnostics	Outcome-based subscription; raised $35M	[54][56]
SKF / Fluke / Emerson	Incumbent instruments + services	Hardware + service contracts	[50]
Component layer (ADI/ST/TI MEMS + MCUs)	Enabling commoditization	Catalog silicon	[48]
Open/academic node designs	Zero-cost reference designs	n/a	[42][48]

2.12 Confidence Score & Evidence Gaps

Confidence: 7/10. The saturation verdict is robust — it rests on two independent result sets, a full-text read of the foundational noise-rejection patent, and convergent academic literature [46][47][118][48].

Evidence gaps: (1) CPC-level filing counts (G01M 13/04, G01H 1/00) were not retrievable in this pass, so saturation is assessed from result-set density rather than bibliometrics; (2) the exact claim scope of Infinite Uptime's patented technology is inferred from press coverage [54][52]; (3) no measured data exists on single-node attribution accuracy in dense shops — the concept's pivotal performance claim remains untested by anyone, including incumbents.

3. TrueMoist — Drift-Self-Correcting Soil Moisture Controller

3.1 Problem Summary

Capacitive (dielectric) soil-moisture sensors are the default low-cost irrigation input, but their raw readings drift with **soil salinity (electrical conductivity), temperature, and aging/corrosion**, so a probe calibrated at installation can be materially wrong months later — driving over- or under-irrigation precisely in the fertigated, high-value crops where accuracy pays [63][64]. The physics is unforgiving: water's relative permittivity (~80) dominates the soil dielectric, but it is itself temperature-dependent, and bulk EC couples into low-frequency capacitance measurements, as the patent record itself explains in detail [115][114]. Peer-reviewed studies repeatedly show that salinity and temperature materially bias capacitive probes unless compensation is applied [63][64][65][66].

The unmet need, as framed in Phase 1C, is a controller that **self-corrects drift on-node** — estimating and compensating salinity/temperature effects continuously, without the user re-calibrating or sending soil samples to a lab. The target users are commercial horticulture and smallholder drip-irrigation operations that cannot justify research-grade probes (METER TEROS-class) but are ruined by hobby-grade drift [59][70].

3.2 Commercial Ecosystem Scan

The market splits into three layers. Research-grade vendors — METER (TEROS dielectric line), Campbell Scientific, Sentek — deliver accurate, temperature/EC-aware instruments at prices that reflect lab positioning [59][60]. Mid-market agronomy platforms — **CropX** being the most visible, with substantial venture backing — bundle decent sensors with analytics and irrigation advice, monetizing the service rather than the probe [70][71]. At the bottom, hobbyist capacitive probes sell for a few dollars and are notorious for corrosion and drift, a failure mode documented across maker and academic evaluations [64][66].

The structural gap is a **mid-price probe with research-grade correction**: vendors at the top embed multi-parameter compensation in expensive instruments; vendors at the bottom ship raw RC-time-constant boards with no compensation at all. CropX-class platforms partially close the gap but lock customers into subscriptions and their own hardware [70][71]. TrueMoist's controller positioning (sensor-agnostic correction + irrigation actuation) would sit between the layers, but the patent record shows the compensation *concept* is far from new — the commercial gap persists because of calibration economics and soil-specificity, not because nobody thought of correcting for EC and temperature.

3.3 Patent & Prior-Art Landscape

The compensation space is well-populated at the circuit and probe level. Read at full text: **US 7,884,620** — "Sensor for measuring moisture and salinity" (dual-frequency bridge: high-frequency capacitance channel for moisture, low-frequency resistance channel for salinity, with ratiometric, self-taring detectors that null temperature drift; priority 2008, expected expiry ~2028) [114]; and **US 11,598,743** — "Soil monitoring sensor including single probe and temperature compensation" (resonant single-probe sensor with an integrated temperature sensor compensating the moisture determination; Korean priority 2020, active to ~2040) [115]. The surrounding result sets add temperature-adjusted probe electronics (US 9,949,450), automatic calibration methods (US 9,585,307), absolute-reading calibration-free claims (US 6,657,443), and a 1974 temperature-and-salinity compensation patent (US 3,782,179, long expired) [68][69]. The '620 patent itself situates US 5,479,104 as prior art on simultaneous capacitance/conductance measurement [114].

Academic prior art is equally thick: temperature and salinity effects on capacitive/TDR probes are a standard study topic with published correction regressions per soil type [63][64][65][66]. Net assessment: **"measure EC and temperature, then correct" is anticipated many times over** at both the circuit and method level. The '743 patent is the sharpest fence — its claims cover a single probe plus temperature sensor plus compensated determination, which reads uncomfortably close to a naive version of TrueMoist [115]. Residual novelty would need to live in the *self*-correcting aspect: on-line, in-situ estimation that tracks drift over seasons without user calibration events, ideally soil-agnostic — a harder, more specific method claim than anything screened in this pass, but not provably clear of the dense Korean/CN filing tail [68][69].

3.4 Technical Prior-Art and Feasibility

A working TrueMoist prototype is straightforward: a capacitive front-end, a bulk-EC channel (two- or four-electrode), a soil-temperature sensor, and a regression/polynomial correction running on any MCU — the '620 patent demonstrates a complete, manufacturable embodiment of exactly this architecture, and its expiry (~2028) will soon free even that specific circuit [114]. The academic literature supplies correction models and quantifies the residual errors [63][64]. Feasibility risk concentrates in **soil-specificity**: dielectric-EC-moisture coupling varies with texture, clay mineralogy, and organic matter, so a single factory calibration cannot be universal; research probes solve this with soil-specific calibration routines that TrueMoist explicitly wants to eliminate [65][66].

The credible technical path is **in-situ self-calibration**: using long-horizon signal structure (diurnal temperature cycles, irrigation/dry-down events) as natural excitation to continuously re-fit correction parameters on-node. This is implementable with known statistics, but its accuracy claim must be validated across soil types — a season-scale field program, not a bench exercise. Feasibility: moderate-to-high for assisted correction; unproven for fully calibration-free operation.

3.5 White-Space Assessment

White space is **low-to-moderate**. Closed zones: dual-frequency moisture+salinity circuits ('620, expiring) [114]; single-probe + temperature compensation ('743, active) [115]; generic auto-calibration and absolute-reading claims [69]. Open or thin zones: (a) **event-driven self-recalibration methods** (using irrigation/dry-down signatures as reference points) — not found in this pass, plausibly novel as a specific method; (b) **soil-agnostic universal correction** with published multi-soil validation — a credibility gap in the whole market [65]; (c) the **controller integration layer** — correction feeding closed-loop irrigation decisions with audit trails, where product (not patent) differentiation lives [70].

The board should treat TrueMoist as a product-engineering play with a possible narrow method filing, not as a platform-IP opportunity.

3.6 Competitive & Defensibility Analysis

Competition: METER/Campbell at the top (accuracy brand), CropX/Sensoterra in the service-bundled middle [70][71], and commodity probes below. A TrueMoist product would fight on "research-grade readings at prosumer prices, no subscription." Defensibility through patents is narrow (see §3.3–3.5); defensibility through **validation data** is more realistic — a published, multi-soil accuracy benchmark would be a genuine market asset because no low-cost vendor has one [64][66]. Distribution through drip-irrigation dealers and government micro-irrigation programs matters more than IP in this category.

A second defensive angle: the correction algorithm is embedded firmware — keep it as a trade secret, publish only the validation results. This avoids disclosing method claims that a Korean/CN filing tail might surround anyway [68][115].

3.7 Regulatory / Standards Context

No device-level mandate exists for soil sensors. Demand-side programs are the relevant context: micro-irrigation subsidies (e.g., India's PMKSY) and water-use-efficiency regulations in water-stressed regions indirectly fund sensor-equipped drip systems; pesticide/fertilizer stewardship programs favor documented irrigation control. Metrological self-declaration (CE) suffices; there is no certification moat either way. The regulatory tailwind is real but soft — it subsidizes the customer's purchase rather than obligating it.

3.8 Risk Matrix

Risk	Likelihood	Impact	Mitigation
'743 single-probe temp-compensation claims read onto product	Medium	High	Claim-chart before design freeze; consider two-probe or method-level design-around [115]
Calibration-free accuracy fails on clay/saline soils	Medium–High	High	Multi-soil field validation as Phase 3 gate [65][66]
'620 expiry (~2028) floods market with free dual-frequency designs	Medium	Medium	Differentiate on self-calibration software + validation data, not circuit [114]
CropX-class platforms add correction as a feature	Medium	High	Sensor-agnostic positioning; dealer channels [70]
Commodity probe price erosion	High	Medium	Sell controller + correction, not bare probes
Long agri sales cycles, no mandate	High	Medium	Ride subsidy programs; co-market with drip OEMs

3.9 Recommended Next Step

Hold pending a validation sprint. Before any filing or product commitment, run a two-season, three-soil-type field trial comparing on-node self-corrected readings against gravimetric truth and a research-grade reference. Advance only if the self-correcting loop demonstrably holds accuracy without user recalibration; if it does, protect the specific event-driven recalibration method (narrow filing) and compete on published accuracy data. If the loop underperforms, fold the concept into a CropX-style service analysis and exit hardware.

3.10 Key Patent & Prior-Art Table

Document	Year	Holder / type	Content	Status & relevance
US 3,782,179	1974	—	Temperature- and salinity-compensated measurement [68]	Expired — concept is 50 years old
US 5,479,104	1995	—	Simultaneous soil capacitance/conductance sensor [114]	Expired — cited as prior art inside '620
US 7,884,620	2011	Campbell-family	Dual-frequency moisture + salinity bridge, self-taring detectors [114]	Expected expiry ~2028 — soon free
US 6,657,443	2003	—	"Absolute reading," no-calibration soil sensor [69]	Likely expired — anticipates calibration-free claims
US 9,585,307	2017	—	Automatic calibration method (optical) [69]	Active — calibration-method fence
US 9,949,450	2018	—	Probe with temperature-adjustment electronics [68]	Active — circuit fence
US 11,598,743	2023	Korean assignee	Single probe + temperature compensation [115]	**Active — sharpest fence**
Salinity/temperature effect studies	2020–2024	Academic	Quantified EC/temperature bias + corrections [63][64][66]	Non-patent prior art

3.11 Commercial Landscape Table

Player	Layer	Positioning	Evidence
METER (TEROS)	Research grade	Accurate dielectric probes, premium pricing	[59][60]
Campbell Scientific / Sentek	Research grade	Full monitoring stations	[68]
CropX	Mid-market platform	Sensor + agronomy SaaS, venture-backed	[70][71]
Sensoterra	Mid-market	LoRa probe fleet	[70]
Commodity capacitive probes	Hobby/OEM	$-class boards, drift/corrosion-prone	[64][66]

3.12 Confidence Score & Evidence Gaps

Confidence: 7/10. Both on-point patents were read in full; the academic drift literature is convergent; commercial layering is well-evidenced [114][115][63].

Evidence gaps: (1) Korean/CN application tail (the '743 family and siblings) not fully enumerated — needed before any filing; (2) no published data on fully calibration-free multi-season accuracy exists anywhere, so the concept's core performance claim is unproven in the open literature [65][66]; (3) mid-market vendor pricing (CropX/Sensoterra per-point costs) is partly opaque — Crunchbase-tier data only [71]; (4) InPASS coverage of Indian agri-sensor filings was out of reach this pass.

4. TrustLatch — Accessible Secure-Boot & Signed-OTA Toolkit for Constrained MCUs

4.1 Problem Summary

Every connected embedded product now needs two security primitives: assurance that only authentic firmware executes (secure boot) and a channel to deliver signed updates that cannot be tampered with in transit or rollback-attacked (signed OTA). The primitives are mature — the open-source MCUboot project documents and implements exactly this chain for 32-bit MCUs [75][79] — yet field evidence and practitioner literature consistently show small teams shipping products without them, because correct integration (key generation, secure storage, image signing, slot management, recovery) is a multi-week specialist effort per MCU family [74][76]. The result is a large population of fielded devices that cannot be patched safely — a systemic vulnerability that regulators have now moved to eliminate (see §4.7).

TrustLatch, as framed in Phase 1C, is a **toolkit** — curated bootloader configurations, key-management workflow, signing service, and reference integrations for constrained targets (ESP32, STM32, nRF52, AVR/PIC-class) — that compresses adoption from weeks to hours. The concept is explicitly accessibility-driven: the invention is not new cryptography but the removal of everything that makes existing cryptography go unused [74][78].

4.2 Commercial & Open-Source Ecosystem Scan

Uniquely among the five concepts, TrustLatch's "competitors" are mostly free. **MCUboot** (Linux Foundation) provides a secure bootloader with image signing and swap/rollback logic and is the default in Zephyr [75][78][79]; **TF-M** provides PSA-certified security services on ARMv8-M; silicon vendors ship their own chains (ESP32 Secure Boot v2, ST SBSFU). On the OTA side, **Mender** anchors the update-server category with an open-source core and commercial SaaS [80][81], flanked by RAUC, SWUpdate, and hawkBit for Linux-class targets, while practitioner guides codify secure-OTA patterns for resource-constrained fleets [74].

Commercial security silicon (secure elements, HSMs) and cloud provisioning services cover the top of the market, and academic work keeps publishing MCU secure-boot frameworks — confirming that the *knowledge* is abundant [76][77]. What nobody sells well is **opinionated, cross-vendor, constrained-MCU onboarding**: the ecosystem is a box of parts. The commercial read: the money is not in the boot chain but in compliance tooling, fleet key management, and update operations — Mender's SaaS trajectory demonstrates willingness to pay for exactly that adjacent value [80][81].

4.3 Patent & Prior-Art Landscape

The patent layer is dense but **mis-aimed** relative to TrustLatch's niche. Read at full text: **US 10,740,084** — SoC-assisted resilient boot, in which an immutable root of trust inside an SoC authenticates and applies firmware updates, with recovery paths; the specification explicitly references Intel CSME-class architecture [119]. **US 9,953,166** — a Microsemi/Microchip-family method for **securely booting processors that lack built-in secure boot**, using an external secure root-of-trust chip that emulates boot memory, serves a nonce, and verifies a challenge–response MAC (with PUF hardware binding and penalty responses); priority 2013, expected active into

~2033 [120]. **US 11,042,609** — secure-element **registration and provisioning** systems for IoT devices (deferred PKI provisioning, certificate monetization at the vendor level) [121]. Result sets add Microsoft/Dell/HP-class filings on secure boot firmware update (e.g., US 8,589,302; US 9,483,246) and hardware-enforced firmware security (US 10,839,080) [82][83].

Three conclusions follow. One: **everything cryptographic is either open source or patented by someone else** — there is no IP to originate in algorithms, chains of trust, or signed-image formats [75][79][119]. Two: the active patents concentrate on richer platforms (SoCs with ROM roots of trust, TPM/secure-element provisioning) — the constrained-MCU, no-hardware-RoT segment is comparatively thin precisely because the open-source stack already serves it, which deters filing [120][83]. Three: the one patent that does read onto the constrained segment — US 9,953,166's external-RoT challenge–response — does not block a *software toolkit* that relies on the MCU's own boot ROM or flash protection (the mainstream ESP32/STM32 path), but it would matter if TrustLatch ever shipped a companion hardware root-of-trust dongle [120]. Prior-art risk to the toolkit concept itself is low because toolkits are product/service constructs, not patent claims.

4.4 Technical Prior-Art and Feasibility

Feasibility is the highest in the portfolio: every component exists, is documented, and has production deployments. MCUboot's docs and repositories provide the bootloader, image format, and signing tools; vendor ROMs provide first-stage verification; Mender/RAUC-class systems provide server-side update operations [75][79][80]. The engineering content of TrustLatch is integration curation — per-MCU reference ports, a key ceremony workflow, CI signing, and recovery drills — plus the documentation and default configurations that determine whether a small team actually finishes the job [74][78].

The real technical risks are adoption-friction details rather than feasibility: key custody on Windows-based factory floors, rollback-counter management across silicon quirks, and secure storage on MCUs without TrustZone. All are solvable with opinionated defaults; none is novel enough to patent. The concept can reach a usable v1 in weeks, and its compliance-mapping artifacts (which CRA/PSTI clauses each feature satisfies) can be authored immediately [74][76].

4.5 White-Space Assessment

White space is **moderate-to-high but non-patentable in kind**. The gap the concept exploits — the distance between "secure boot exists" and "a two-person team actually ships it" — is visible across practitioner literature and open-source issue trackers [74][78]. It is not filled by patents because it is a usability/services gap; incumbents ignore it because enterprise security vendors monetize at price points inaccessible to the long tail, and open-source projects deliberately stop at reference code [75][80].

The thin patent zones that do exist are peripheral: factory key-provisioning workflows for secure-element-less MCUs (compare US 11,042,609's provisioning focus, aimed at SE-equipped devices) [121], and compliance-automation tooling (mapping firmware-update capabilities to regulatory clauses). The board should not expect a patent moat here; the realistic protection is **brand, certification-mapping content, and network effects** in an open-core model.

4.6 Competitive & Defensibility Analysis

Competition comes from four directions: do-it-yourself open source (MCUboot + Mender CE, free but unopinionated) [75][80]; silicon-vendor tooling (free but single-vendor and legally non-committal); enterprise security platforms (comprehensive, enterprise-priced); and internal security teams at larger OEMs (who will never buy). TrustLatch's wedge is the cross-vendor long tail with a compliance deadline — precisely the customers the first three options fail [74][76].

Defensibility: weak in patents, potentially strong in **trust assets** — audited reference ports, a compliance-mapping knowledge base, signed-release provenance, and community adoption. These compound with usage and are hard for a vendor tool to copy credibly. The strategic posture should be open-core: give away the per-MCU integrations (maximize adoption and trust), monetize fleet key management, update operations, and compliance reporting.

4.7 Regulatory / Standards Context

This is TrustLatch's decisive advantage: it is the only portfolio concept with a **hard regulatory clock**. The EU Cyber Resilience Act obligates manufacturers of products with digital elements to provide secure update mechanisms and vulnerability handling, with penalties on turnover; the UK PSTI Act already bans universal default passwords and requires vulnerability disclosure; US federal procurement flows through NISTIR 8259 / the IoT Cybersecurity Improvement Act; ETSI EN 303 645 and IEC 62443 provide the harmonized technical baselines — all converging on "secure boot + signed update capability" as table stakes [74][76]. Practitioner coverage frames secure OTA explicitly as the compliance mechanism, not merely good hygiene [74].

Every one of these instruments converts TrustLatch's value proposition from optional to obligatory for exactly the resource-constrained manufacturers least able to self-implement [76][77]. The regulatory pull score for this concept is the portfolio maximum (9/10).

4.8 Risk Matrix

Risk	Likelihood	Impact	Mitigation
Open-source projects ship "good-enough" opinionated tooling, erasing the wedge	Medium	High	Contribute upstream; monetize operations/compliance, not the boot chain [75][79]
Silicon vendors bundle free toolkits to sell chips	Medium	High	Stay cross-vendor; target multi-silicon product companies
No patent moat → easy imitation	High	Medium	Accept; build brand/audit/compliance assets instead
External-RoT patent exposure if hardware dongle pursued later	Low–Med	Medium	Avoid companion-RoT hardware or license '166-family [120]
Security defect in the toolkit itself destroys trust	Low	Existential	Third-party audits, signed releases, reproducible builds
CRA enforcement slower than expected delays urgency	Medium	Medium	Sell PSTI/NIST procurement hooks in parallel [74][76]

4.9 Recommended Next Step

Advance immediately — as an open-core product, not a patent program. Ship v1 covering three MCU families (ESP32, STM32, nRF52) with one-command provisioning, CI signing, and a CRA/PSTI compliance-mapping whitepaper; price fleet key management and update operations as SaaS. Do not spend on filings beyond trademark. Use VibeGuard-style connected-sensor products as first design partners — the portfolio's own concepts are reference customers [74][75].

4.10 Key Patent & Prior-Art Table

Document	Year	Holder / type	Content	Status & relevance
US 9,953,166	2018	Microsemi/Microchip-family	External secure RoT boots non-secure processors; challenge–response MAC + PUF binding [120]	**Active (~2033) — constrains hardware-RoT variants**
US 10,740,084	2020	Major SoC vendor	In-SoC immutable RoT authenticates/applies firmware updates; recovery [119]	Active — rich-platform fence
US 11,042,609	2021	Industrial assignee	Secure-element registration & deferred PKI provisioning for IoT [121]	Active — provisioning fence
US 8,589,302 / US 9,483,246	2013/2016	PC-industry assignees	Secure boot / firmware update [82]	PC/server-oriented
US 10,839,080	2020	—	Hardware-enforced firmware security [82]	Active — platform fence
MCUboot (docs + repo)	2016–	Linux Foundation	Open secure bootloader, signing, slots, rollback [75][79]	Free — the toolkit's substrate
Mender (repo + SaaS)	2016–	Northern.tech	Open-core OTA update server [80][81]	Free core — business-model template
MCU secure-boot academic frameworks	2020–2025	Academic	Constrained-device boot frameworks [76][77]	Non-patent prior art

4.11 Commercial & OSS Landscape Table

Player	Layer	Model	Evidence
MCUboot / Zephyr	Bootloader	Open source (Linux Foundation)	[75][78][79]
Mender	OTA server	Open core + SaaS	[80][81]
RAUC / SWUpdate / hawkBit	Linux-class OTA	Open source	[74]
Silicon vendors (ESP, ST, NXP, Microchip)	ROM boot + vendor tools	Free with silicon	[74][120]
Secure-element/PKI provisioning vendors	RoT hardware + provisioning	Per-unit + service	[121][83]

Enterprise security platforms	End-to-end device security	Enterprise license	[82]

4.12 Confidence Score & Evidence Gaps

Confidence: 8/10. The open-source substrate, the compliance calendar, and the three anchor patents were all verified at primary-source level; the "usability gap" premise is corroborated across practitioner and academic sources [74][75][76][120].

Evidence gaps: (1) CRA delegated-act details (exact product-scope thresholds and dates) were not re-verified this pass and should be pinned before go-to-market claims; (2) Mender's current pricing/ARPU is inferred from its model, not from financials [81]; (3) no measured baseline exists for "weeks-to-hours" integration effort — the core value metric needs a benchmark study; (4) the '166 patent family continuations (Microchip) should be enumerated if any hardware-RoT roadmap emerges [120].

5. ColdTrace — Multi-Modal Cold-Chain Integrity Logger

5.1 Problem Summary

Temperature-sensitive products — vaccines first among them — are damaged by both heat **and freezing** during storage and transport, and the damage is often invisible at the point of care. A systematic literature review found freezing exposure in **14–35% of vaccine shipments** studied, with the problem persisting in a 2018 follow-up review [93][95][97]; WHO immunization guidance and industry whitepapers treat excursion management as a core cold-chain discipline [100][92]. The monitoring status quo has two failure modes that Phase 1C targeted. First, **alarm fatigue**: loggers that record air temperature inside a shipper alarm on every door opening or defrost cycle because air responds in seconds while the product's thermal mass moves in hours — pharmacy and vaccine-monitoring guidance consequently recommends *buffered* probes that approximate product temperature [90][91][96][98]. Second, **single-modality**: temperature loggers see thermal history but not handling; shock/tilt indicators see handling but not potency-relevant thermal dose — and vibration/handling stress matters for sensitive biologics and food quality alike [102][103].

ColdTrace, as framed in Phase 1C, is a multi-modal logger — temperature modeled against product (not air) response, plus shock/tilt/light context — delivering excursion *classification* ("harmless door opening" vs "product-threatening event") rather than raw threshold alarms. The evidence for need is strong and programmatic (WHO/CDC/GDP), not merely anecdotal [93][100][92].

5.2 Commercial Ecosystem Scan

The incumbent stack is layered and surprisingly cheap at the bottom: chemical time-temperature indicators (TTIs) and vaccine vial monitors cost cents-to-dollars per unit and are mandated on many vaccine shipments [116]; single-use USB/PDF loggers and 30-day electronic temperature recorders serve shipment and storage roles [88][96]; reusable multi-use loggers and pharmacy monitoring systems serve the CDC VFC-program storage segment, where buffered probes are standard practice [96][98]. Condition-indicator specialists — **SpotSee** (ShockWatch/ShockLog/SpotBot brands) — sell shock, tilt, and temperature indicators as parallel product families across pharma, food, and electronics logistics [102][103][104], and real-time cellular trackers occupy the premium pallet tier [88].

The structural gap: each layer answers a different question (Was it ever too hot? Was it dropped? What is the temperature now?), and none answers ColdTrace's question — *"did this specific excursion actually threaten the product, and what should I do?"* — at commodity prices. Excursion intelligence exists today only inside expensive real-time platforms or as human interpretation of downloaded traces [91][98]. The gap is real, but the patent scan below shows two active fences around the most obvious implementations.

5.3 Patent & Prior-Art Landscape

Read at full text, four documents define the terrain. **US 7,102,526** — "Electronic time-temperature indicator and logger" (2003 priority): the canonical **stability-bank** algorithm, integrating temperature-dependent degradation P(T) against a budget F to compute remaining shelf life, with explicit discussion of non-Arrhenius profiles, freeze damage, and phase-transition penalties; it also surveys the chemical TTI art (3M MonitorMark US 5,667,303; TempTime Heatmarker) and the Sensitech TagAlert alarm (US RE 36,200) [116]. With 2003 effective priority, this foundational algorithmic-TTI patent is **expected expired (~2023–24)** — freeing model-based excursion scoring as a design approach [116]. **US 10,887,735** — "Vaccine monitoring system" (priority 2018, granted 2021): an insulated container with a **buffered temperature sensor chamber whose buffering agent "simulates the physics that the vaccines undergo"**, plus cellular alerting, 2–8 °C notification ranges, and rate-of-change logic that predicts whether the shipment will arrive before limits are exceeded [117]. This is the **single most on-point active patent against ColdTrace**: the thermal-mass/buffered-probe mechanism that solves alarm fatigue is claimed here in a vaccine-carrier embodiment, active to roughly 2038 [117].

The surrounding result sets reinforce the picture: recent pallet/container monitoring patents (US 12,248,900; US 12,524,729 — 2024–25 filings showing continued activity), cellular cold-chain logistics families (US 10,592,849), temperature-limit indicators (US 11,467,042), and vaccine monitoring variants (US 8,671,871) [88][89]. Conclusion: (a) **algorithmic excursion scoring is now free** (expired '526) — a genuine white-space gift; (b) **physical buffered-probe implementations for vaccine carriers are fenced** by '735 — ColdTrace must implement product-temperature estimation *differently* (e.g., a two-sensor air+thermal-model approach — a virtual thermal mass computed in firmware — rather than a physical buffer chamber), and obtain a formal FTO opinion on '735's claim scope [117]; (c) multi-modal shock+temp fusion at the logger level appears in indicator-brand product lines but is thinner in the patent record than cellular pallet tracking, suggesting room for specific fusion/claims [88][102].

5.4 Technical Prior-Art and Feasibility

Feasibility is high. The sensing set (NTC/digital temperature, MEMS accelerometer for shock/tilt, light sensor, coin-cell or $LiSOCl_2$ power, local logging plus BLE/NFC readout) is commodity engineering, and the expired '526 patent supplies a complete, validated algorithmic framework — stability banking with arbitrary P(T) tables — that can be implemented in a few hundred lines of firmware [116]. The alarm-fatigue solution has two technical routes with different IP exposure: the *physical* buffered probe (standard pharmacy practice [96][98], but fenced for vaccine carriers by '735 [117]) versus a *virtual* thermal mass — estimating product temperature from air-temperature dynamics via a first-order thermal model, computable on any MCU. The virtual route is standard control-systems engineering and, on this record, does not appear claimed for the excursion-classification use-case.

The differentiating intelligence — classifying excursions by integrating thermal dose, rate-of-change, shock events, and lid-open (light) events into an actionable verdict — is an embedded-analytics problem with no physics barrier. Feasibility risk is mostly **product**: hitting 30–60-day battery life, unit-cost targets competitive with single-use loggers, and regulatory documentation (21 CFR Part 11-style records) that pharma customers demand [98][90].

5.5 White-Space Assessment

White space is **moderate**, and newly improved by an expiry. Open zones: (a) **virtual-thermal-mass excursion classification** — model-based product-temperature estimation with multi-modal context, distinct from '735's physical buffer [117]; (b) the expired stability-bank algorithm as a free algorithmic base for potency-aware scoring [116]; (c) **low-cost multi-modal fusion** (temperature + shock + light) positioned below real-time cellular trackers — a price/feature tier the indicator brands (discrete, single-modality) and tracker platforms (expensive, subscription) both miss [102][88]. Closed or fenced zones: physical buffered-chamber vaccine carriers ('735) [117]; cellular real-time pallet logistics (dense, active) [88]; chemical TTI chemistry (3M/TempTime families) [116].

The white-space verdict: pursue the *intelligence* layer (classification, virtual mass, fusion) at commodity hardware prices; avoid claiming buffered probes and avoid the cellular-tracker business.

5.6 Competitive & Defensibility Analysis

Competitors by layer: chemical TTI/VVM suppliers (cents per unit, mandated — coexistence, not competition); single-use logger brands (DeltaTrak/Sensitech-class) and pharmacy monitoring systems serving VFC storage [96][88]; SpotSee's indicator families on the shock/condition side [102][104]; and real-time tracker platforms above. ColdTrace's wedge is the **interpretation gap** between these layers: it sells fewer false alarms and a defensible disposition record.

Defensibility: one possible narrow patent position exists — the specific multi-modal excursion-classification method (virtual thermal mass + shock/light context + stability scoring) — subject to FTO review against '735 and the indicator-brand portfolios [117][102]. Stronger protection is again evidentiary: published false-alarm-reduction benchmarks against standard air-temperature loggers would be a marketable, hard-to-copy asset, and 21 CFR Part 11-style audit trails create switching costs in pharma accounts [98][90].

5.7 Regulatory / Standards Context

Demand-side regulation is strong and specific. WHO Performance, Quality and Safety (PQS) specifications govern vaccine cold-chain devices; CDC's Vaccines for Children program requires calibrated continuous temperature monitoring with buffered probes in storage units [96][98]; WHO immunization handbooks operationalize excursion handling [100]; EU GDP and 21 CFR Part 11 frame pharma distribution records and electronic data integrity; FSMA covers food. Freezing — the under-recognized killer — is explicitly documented by the WHO-cited systematic review at 14–35% shipment exposure [93][95].

These frameworks do not mandate ColdTrace's multi-modal classification, but they (i) mandate the monitoring layer it attaches to, (ii) increasingly demand defensible excursion disposition (exactly the product), and (iii) the VFC buffered-probe norm validates the product-temperature logic at the concept's core [96][98]. Regulatory pull: 8/10 — second only to TrustLatch.

5.8 Risk Matrix

Risk	Likelihood	Impact	Mitigation
US 10,887,735 claims read onto buffered/thermal-mass implementation	Medium–High	High	Use virtual (model-based) thermal mass; formal FTO opinion pre-launch [117]
Incumbent logger brands add excursion-classification firmware	Medium	High	Ship benchmark data fast; lock pharma accounts with audit-trail feature
Price floor set by single-use loggers compresses margin	High	Medium	Multi-use design + service record; target storage + last-mile, not single-use freight

Validation burden (multi-product stability tables) slows launch	Medium	Medium	Start with 2–8 °C vaccine profile + one food profile using published stability data [116][93]
Regulatory documentation underestimated (Part 11, PQS)	Medium	High	Budget compliance engineering from day one [98]
Chemical TTI/VVM incumbency at vial level	Low–Med	Low	Position at shipper/storage level, complementary to VVM [116]

5.9 Recommended Next Step

Advance with conditions. Proceed to Phase 3 prototype on the virtual-thermal-mass + multi-modal classification architecture, gated on (i) a formal FTO opinion on US 10,887,735 and the indicator-brand portfolios, and (ii) a bench validation showing false-alarm reduction versus an air-temperature logger on standardized door-opening/defrost profiles [91][96]. File narrowly on the classification method if FTO clears; anchor go-to-market on VFC storage monitoring and vaccine last-mile carriers where freezing risk (14–35% exposure) is documented [93][96].

5.10 Key Patent & Prior-Art Table

Document	Year	Holder / type	Content	Status & relevance
US 7,102,526	2006 (2003 priority)	TTI algorithm	Electronic TTI/logger with stability-bank P(T) model; surveys 3M/TempTime/Sensitech art [116]	**Expected expired — free algorithmic base**
US RE 36,200 (Sensitech TagAlert)	reissue	Sensitech	Threshold temperature alarm [116]	Expired — background
US 5,667,303 (3M MonitorMark)	1997	3M	Wicking chemical TTI [116]	Expired — background
US 10,887,735	2021 (2018 priority)	Vaccine monitoring system	Buffered-sensor chamber simulating vaccine thermal physics + cellular alerting + rate-of-change [117]	**Active (~2038) — primary FTO fence**
US 8,671,871	2014	—	Vaccine monitoring variant [89]	Active — secondary fence
US 10,592,849	2020	—	Cellular cold-chain logistics [88]	Active — tracker tier
US 12,248,900 / US 12,524,729	2024–25	—	Pallet/container monitoring	Active — filing velocity signal [88]
Matthias et al., Vaccine	2007; 2018 follow-up	Peer-reviewed	Freezing exposure in 14–35% of shipments [93][95][97]	Key demand evidence

5.11 Commercial Landscape Table

Player	Layer	Positioning	Evidence
Chemical TTI / VVM suppliers	Vial/unit level	Cents-per-unit, mandated on many shipments	[116]

Single-use & 30-day loggers (DeltaTrak/Sensitech-class)	Shipment level	Commodity hardware	[88][96]
Pharmacy/VFC monitoring systems	Storage level	Buffered probes standard; compliance-driven	[96][98][90]
SpotSee (ShockWatch/ShockLog/SpotBot)	Condition indicators	Shock/tilt/temperature indicator families	[102][103][104]
Real-time cellular trackers	Premium pallet tier	Hardware + subscription	[88]
SenseAnywhere-type guidance vendors	Excursion practice	Alarm-fatigue discourse validates need	[91]

5.12 Confidence Score & Evidence Gaps

Confidence: 7/10. The two decisive patents ('526 expired base, '735 active fence) were read in full; the freezing-exposure statistics come from the canonical peer-reviewed review; the buffered-probe norm is documented across CDC-facing guidance [116][117][93][96].

Evidence gaps: (1) '735's continuation family and any design-around case law not reviewed — requires counsel; (2) unit economics of single-use loggers (true street prices) were not retrievable this pass, so the price-tier gap is characterized qualitatively; (3) shock-damage dose-response data for specific vaccines/foods is thin in open literature, weakening the multi-modal value claim quantitatively; (4) WHO PQS device-category specifics (E006) should be pinned before regulatory marketing claims [100].

6. Comparative Portfolio Analysis

6.1 Cross-Concept Evidence Matrix

The table below aggregates the twelve-section findings into one decision surface. "Free base" identifies lapsed patents each concept may build on; "Active fence" names the document most likely to require design-around or licensing.

Dimension	OpenBraille	VibeGuard	TrueMoist	TrustLatch	ColdTrace
Incumbent price pain	$3.5k–15k displays [2]	Enterprise-tier PdM pricing [43]	Research-grade probes premium [59]	Weeks of specialist effort [74]	False alarms / freezing losses [91][93]
Disruption proof in market	Orbit $449; Braille Me $515.50 [107][110]	Infinite Uptime $35M raise [56]	CropX venture traction [71]	Mender SaaS model [80]	VFC buffered-probe norm [96]
Free base (expired art)	SMA '721, EAP '063 [27][26]	Navy '349 (near expiry) [118]	'179, '104, '620 (~2028) [114]	MCUboot/TF-M (open) [75]	Stability-bank '526 [116]
Active fence	Dot estate; US 11,410,574 [19][29]	Dense platform + CNN filings [46][47]	US 11,598,743 [115]	US 9,953,166 (hardware only) [120]	US 10,887,735 [117]
Patent saturation	Moderate	**High**	Moderate	Moderate (mis-aimed)	Moderate–High
White space	Moderate–High	Low	Low–Moderate	Moderate–High (non-patent)	Moderate
Feasibility	High (market-proven)	High (undifferentiated)	Moderate–High	**Highest**	High
Regulatory pull	7/10	4/10	5/10	**9/10**	8/10
Defensibility path	Mechanism + manufacturing IP	Deployment economics only	Validation data + trade secret	Brand/community (open-core)	Narrow method + benchmark data
Confidence	8/10	7/10	7/10	8/10	7/10

6.2 Portfolio Maps

Figure 1 plots each concept on patent saturation versus white-space opportunity; Figure 2 shows the six-dimension score profile; Figure 3 anchors the OpenBraille analysis in verified market prices — the strongest single piece of evidence in the portfolio that a 5–10× price disruption is executable.

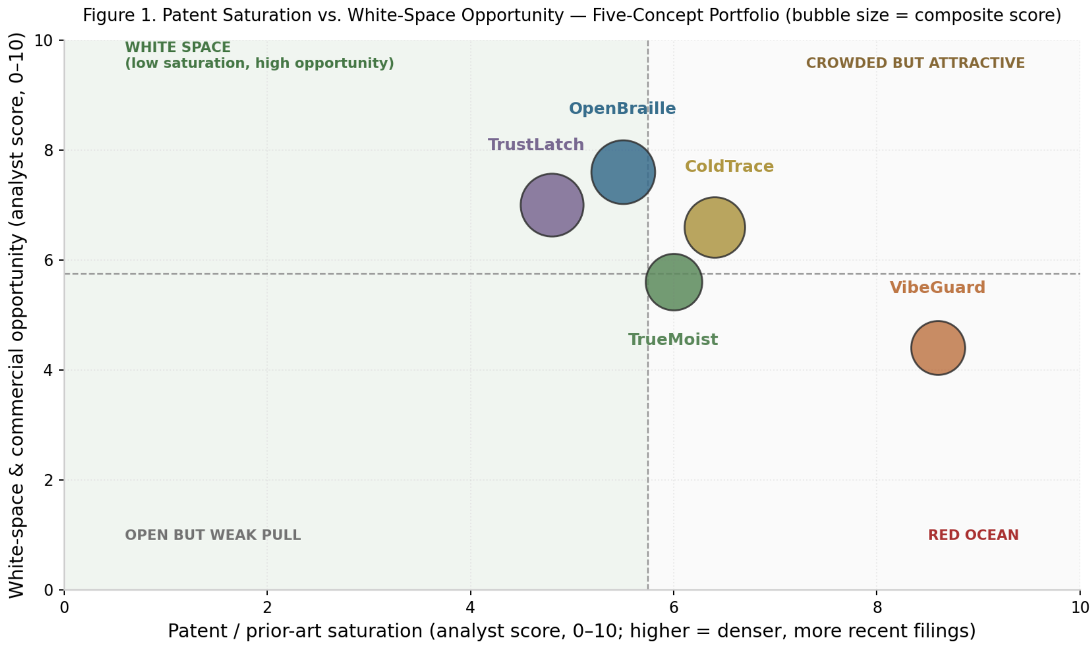

Figure 1. Patent saturation vs. white-space opportunity for the five concepts (analyst scores; bubble size = composite).

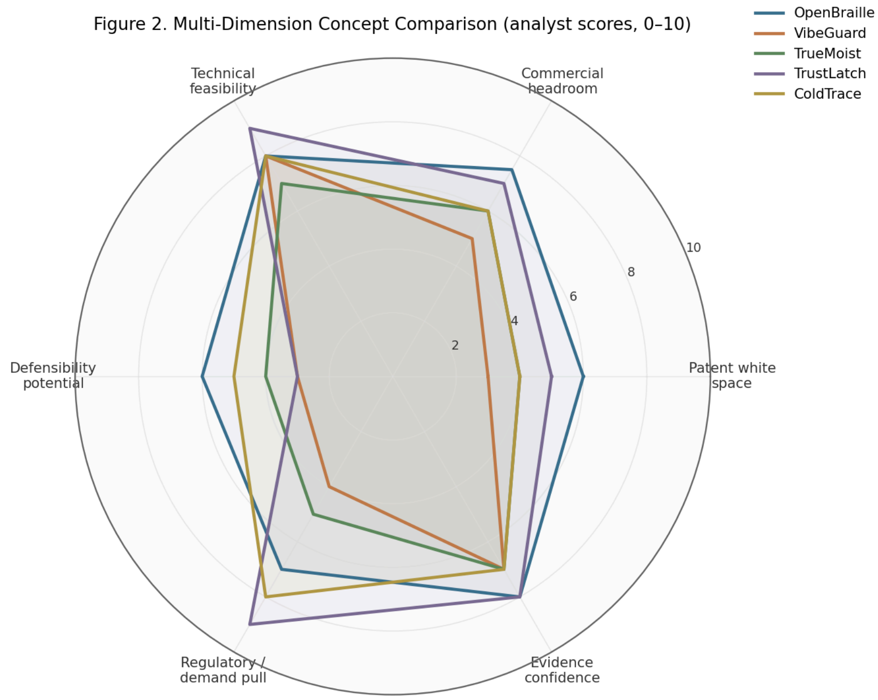

Figure 2. Six-dimension radar comparison (analyst scores, 0–10).

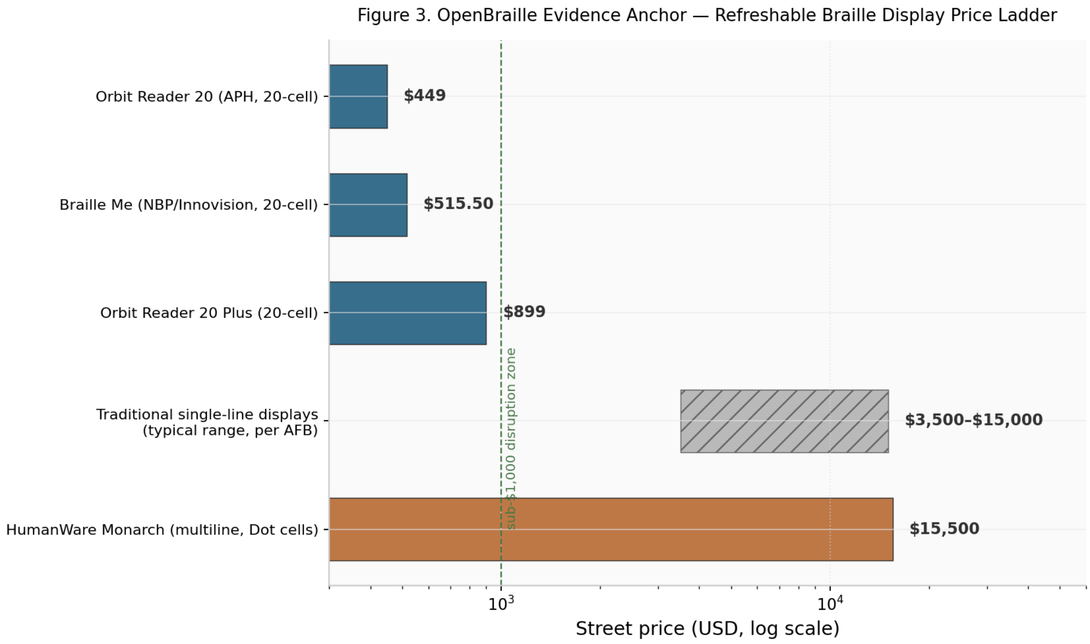

Figure 3. Refreshable braille display price ladder (log scale) — verified street prices against the incumbent band.

6.3 Scoring Rationale

Scores are equal-weighted across six dimensions (patent white space, commercial headroom, technical feasibility, defensibility potential, regulatory/demand pull, evidence confidence), each on a 0–10 ordinal scale justified in the concept chapters. **OpenBraille (7.0)** wins on the combination of a verified $449 market proof against a $3.5k–15k incumbent band, an expired mechanism base to build on, and identifiable residual novelty in manufacturing methods [107][2][27]. **TrustLatch (6.8)** scores highest on feasibility and regulatory pull but is structurally capped on defensibility — its ranking reflects a product opportunity, not an IP opportunity [75][74]. **ColdTrace (6.3)** would rank second on need alone (documented 14–35% freezing exposure) but carries the portfolio's single sharpest active-patent fence [93][117]. **TrueMoist (5.5)** has a real problem with partially fenced solutions and an unproven core performance claim [115][65]. **VibeGuard (5.0)** combines the highest patent saturation with the weakest differentiation — strong technology, wrong battlefield [46][118].

Sensitivity: the ranking of the top two is stable (both lead by ≥0.5 over third place under any plausible weighting); ColdTrace and TrustLatch swap positions if defensibility is weighted above regulatory pull; TrueMoist rises above ColdTrace only if calibration-free field performance validates.

7. Final Ranking & Portfolio Recommendations

Rank	Concept	Composite	Decision	First gate
1	**OpenBraille**	7.0	Advance — patent-track prototype	10^7-cycle endurance + dot-force parity at <$15 cell cost
2	**TrustLatch**	6.8	Advance — open-core product, no filings	v1 on 3 MCU families + compliance whitepaper
3	**ColdTrace**	6.3	Conditional advance	FTO opinion on US 10,887,735 + bench false-alarm benchmark [117]
4	**TrueMoist**	5.5	Hold	Two-season, three-soil validation of self-correction
5	**VibeGuard**	5.0	Demote	Re-scope to niche or exit

Three portfolio-level recommendations. First, **sequence the two advances in parallel**: OpenBraille consumes hardware-engineering resources, TrustLatch consumes firmware/community resources — they do not contend, and TrustLatch's compliance whitepaper doubles as the security foundation for every connected product this pipeline may ship. Second, **run ColdTrace's FTO review immediately**, because its result determines whether the concept is third or drops below TrueMoist: if '735's claims reach virtual thermal-mass estimation, the concept's differentiation collapses into a commodity-logger play [117]. Third, **harvest VibeGuard rather than killing it outright**: its security requirements make it TrustLatch's first internal design partner, and its field-trial data on single-node attribution would be a publishable benchmark with option value.

8. Evidence Gaps & Phase 3 Hooks

The highest-value evidence gaps, in priority order: (1) **Claim-charting of Dot Incorporation's actuator estate** (blocks OpenBraille multiline work) and of US 10,887,735 (blocks ColdTrace's architecture choice) — both are counsel-level tasks that should be commissioned before Phase 3 spending [19][117]. (2) **InPASS and CNIPA coverage** — Indian and Chinese filings in braille cells, soil sensors, and vibration diagnostics were outside this pass; given that two key competitors (Innovision, Infinite Uptime) are Indian and the CNN-diagnosis filing tail is largely Chinese, this gap is material to three of five concepts [34][54][46]. (3) **Unit economics** — Orbit/Braille Me cell costs, single-use logger street prices, and CropX/Sensoterra per-point pricing are inferred rather than measured [110][88][71]. (4) **Performance benchmarks that do not yet exist anywhere**: calibration-free multi-season soil accuracy, single-node vibration attribution in dense shops, and virtual-thermal-mass false-alarm reduction — each is a Phase 3 experiment whose result, positive or negative, is decision-grade [65][48][91].

Phase 3 should therefore open with two parallel tracks: a legal track (claim charts, InPASS/CNIPA sweep, FTO opinions on the two named fences) and an engineering track (OpenBraille endurance gate, TrustLatch v1, ColdTrace bench benchmark). The Phase 2 verdict for the board: **advance OpenBraille and TrustLatch now, gate ColdTrace on counsel, hold TrueMoist for field data, and demote VibeGuard to harvest status.**

This report is for general informational and internal decision-support purposes only. It does not constitute legal advice, a freedom-to-operate opinion, or a patentability determination; patent status and expiry estimates must be verified by qualified counsel before reliance.

Annex A — References

Bracketed numbers in the text correspond to the sources below. Patent records were consulted at full-text level via the FreePatentsOnline mirror of USPTO publications. Result-set entries represent screened candidate-document lists; individually read documents are marked “full text.”

[1] "This $500 Tablet Brings Words to Blind Users’ Fingertips," MIT Technology Review (2017). https://www.technologyreview.com/2017/01/12/243813/this-500-tablet-brings-words-to-blind-users-fingertips/

[2] "Refreshable Braille Displays," American Foundation for the Blind (AFB) product guide. https://afb.org/blindness-and-low-vision/using-technology/assistive-technology-products/refreshable-braille

[3] "Text to Braille Scanner with Ultra Low Cost Refreshable Braille Display," ResearchGate. https://www.researchgate.net/publication/330246078_Text_to_Braille_Scanner_with_Ultra_Low_Cost_Refreshable_Braille_Display

[4] "Low-Cost Solution for Accessible Braille Displays," ISEF project EBED033T. https://isef.net/project/ebed033t-low-cost-solution-for-accessible-braille-displays

[5] "BLITAB: Android Tablet with 14-Row Braille Display," Perkins School for the Blind. https://www.perkins.org/resource/blitab-android-tablet-14-row-braille-display/

[6] Braille display design entry, Create the Future Design Contest (Tech Briefs, 2020). https://contest.techbriefs.com/2020/entries/consumer-product-design/10450

[8] Community discussion of braille display pricing, Reddit r/Blind. https://www.reddit.com/r/Blind/comments/1jztglk/expensive/

[9] "Affordable Braille Displays," Foundation for Blind Children. https://seeitourway.org/affordable-braille-displays/

[11] Low-cost piezoelectric braille display analysis, JIC 2024 poster (Mehta, Y.). https://sspcdn.blob.core.windows.net/files/Documents/SEP/JIC/2024/Posters/2024_JIC_Mehta_Yash_Poster.pdf

[14] "Spotlighting Multi-Line Displays," AFB AccessWorld. https://afb.org/blog/entry/spotlighting-multi-line-displays

[15] HumanWare Monarch product listing, Sterling Adaptives. https://sterlingadaptives.com/products/humanware-monarch?srsltid=AfmBOooSVqGySaj9Y7Gl41BlmDaw09JElL9X4vWAXyEPjXxzjdvE5prU

[16] "Innovative Dot Pad from South Korea Makes Images Touchable," Branding in Asia. https://www.brandinginasia.com/innovative-dot-pad-from-south-korea-makes-images-touchable-for-the-visually-impaired/

[17] "Dot Pad Picks Up Fast Company World-Changing Idea Accolade," LBBOnline. https://lbbonline.com/news/dot-pad-picks-up-fast-company-world-changing-idea-accolade

[19] Dot Pad X product page, Dot Incorporation. https://www.dotincorp.com/en/product/dotpadx

[20] Monarch product page, American Printing House (APH). https://www.aph.org/product/monarch/

[21] Pin-array product architecture survey, ACM Trans. on Computer-Human Interaction (DOI 10.1145/3626468). https://dl.acm.org/doi/pdf/10.1145/3626468

[22] Multi-line tactile graphics display discussion, AppleVis forum. https://www.applevis.com/forum/assistive-technology/discussion-about-top-multi-line-tactile-graphics-displays-out-there

[25] US patent result set: "refreshable braille display actuator," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=refreshable+braille+display+actuator&submit=&patents_us=on

[26] US 6,881,063 — Electroactive polymer actuator braille cell (full text, FreePatentsOnline). https://www.freepatentsonline.com/6881063.html

[27] US 5,685,721 — Refreshable braille-cell display implemented with shape memory alloys (full text). https://www.freepatentsonline.com/5685721.html

[28] US 2024/0363022 A1 — Refreshable braille display system (full text). https://www.freepatentsonline.com/y2024/0363022.html

[29] US 11,410,574 — Layered electromagnetic braille display (full text, FreePatentsOnline). https://www.freepatentsonline.com/11410574.html

[33] "Ultra-Low-Cost, Low-Power, High-Speed Electronic Braille Device," M.Z.A. Mia. https://zesun33.github.io/assets/pdf/Ultra-Low-Cost-Low-Power-High-Speed-Electronic-Braille-Device-for-Visually-Impaired-People-preprint.pdf

[34] Braille Me technology page, National Braille Press (NBP). https://www.nbp.org/ic/nbp/technology/brailleme.html?from_search=1

[36] "Bristol Braille Canute Multi-Line Refreshable Braille Display," Perkins. https://www.perkins.org/resource/bristol-braille-canute-multi-line-refreshable-braille/

[37] "Canute Scientific," Crowd Supply. https://www.crowdsupply.com/bristol-braille-technology/canute-scientific

[38] "Canute: the world's first working refreshable multiline braille display," EMF2016 talk, media.ccc.de. https://media.ccc.de/v/emf2016-176-canute-the-worlds-first-working-refreshable-multiline-braille-display

[39] "About Canute," Bristol Braille Technology. https://bristolbraille.org/about-canute/

[42] Journal of Vibration (MDPI) — vibration engineering research index. https://www.mdpi.com/journal/vibration

[43] Augury corporate site — machine and process health platform. https://www.augury.com/

[44] Augury — About page. https://www.augury.com/about/

[45] Augury peer reviews, Gartner. https://www.gartner.com/reviews/product/augury

[46] US patent result set: "vibration monitoring bearing fault diagnosis," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=vibration+monitoring+bearing+fault+diagnosis&submit=&patents_us=on

[47] US patent result set: "predictive maintenance + vibration + edge + machine learning," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=%22predictive+maintenance%22+vibration+edge+machine+learning&submit=&patents_us=on

[48] Noise-robust bearing fault diagnosis study, MDPI Sensors 23(4):1902. https://www.mdpi.com/1424-8220/23/4/1902

[49] Vibration/fault-diagnosis research article, Extrica. https://www.extrica.com/article/21928

[50] Condition-monitoring techniques and vendor survey, Brunel University research archive (thesis). https://bura.brunel.ac.uk/bitstream/2438/31919/1/FulltextThesis.pdf

[51] Tractian corporate site. https://tractian.com/en

[52] Infinite Uptime directory profile, Industrial Automation India. https://www.industrialautomationindia.in/directory/infinite-uptime

[54] "Infinite Uptime Redefines the Predictive Maintenance…" (press release), Business Standard. https://www.business-standard.com/content/press-releases-ani/infinite-uptime-redefines-the-predictive-maintenance-category-121042800696_1.html

[55] Infinite Uptime corporate site. https://www.infinite-uptime.com/

[56] "Infinite Uptime bags $35M…," Yahoo Finance (AU). https://au.finance.yahoo.com/news/infinite-uptime-bags-35m-help-033028049.html

[59] TEROS soil-sensor line (METER Group), teros.ai. https://teros.ai/en

[60] teros.dev — https://teros.dev/. https://teros.dev/

[63] Salinity/temperature effects on capacitive soil-moisture sensors, SciELO – Revista Caatinga. https://www.scielo.br/j/rca/a/XhFvbZGrWy7Z4NYGWMK9hRn/?lang=en

[64] Soil-moisture sensor temperature/salinity study, MDPI Sensors 20(24):7041. https://www.mdpi.com/1424-8220/20/24/7041

[65] Soil-moisture sensing review, JSAN 13(1):32 (HAL open archive). https://hal.science/hal-04873514v1/file/jsan-13-00032.pdf

[66] Capacitive soil-sensor evaluation study, PMC11478466. https://pmc.ncbi.nlm.nih.gov/articles/PMC11478466/

[68] US patent result set: "soil moisture sensor temperature compensation salinity," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=soil+moisture+sensor+temperature+compensation+salinity&submit=&patents_us=on

[69] US patent result set: "soil moisture sensor calibration correction conductivity," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=soil+moisture+sensor+calibration+correction+conductivity&submit=&patents_us=on

[70] CropX corporate site. https://cropx.com/

[71] CropX profile, Crunchbase. https://www.crunchbase.com/organization/cropx

[74] "Secure OTA Updates & IoT Firmware," Platanor Technologies. https://www.platanor.com/blog/secure-ota-updates-iot-firmware

[75] MCUboot documentation (docs.mcuboot.com). https://docs.mcuboot.com/

[76] IoT device security / secure-boot study, PMC12987191. https://pmc.ncbi.nlm.nih.gov/articles/PMC12987191/

[77] Secure boot on constrained devices, BME CrySyS thesis. https://static.crysys.hu/publications/files/setit/thesis_bme_Gedeon20bsc.pdf

[78] "MCUboot: Secure Boot in Zephyr," DEV Community. https://dev.to/ripan030/mcuboot-secure-boot-in-zephyr-4aoi

[79] GitHub — mcu-tools/mcuboot. https://github.com/mcu-tools/mcuboot

[80] GitHub — mendersoftware/mender. https://github.com/mendersoftware/mender

[81] Mender.io — OTA update platform. https://mender.io/

[82] US patent result set: "secure boot firmware update microcontroller root of trust," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=secure+boot+firmware+update+microcontroller+root+of+trust&submit=&patents_us=on

[83] US patent result set: "key provisioning microcontroller secure element IoT," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=key+provisioning+microcontroller+secure+element+IoT&submit=&patents_us=on

[88] US patent result set: "cold chain temperature monitoring excursion logger," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=cold+chain+temperature+monitoring+excursion+logger&submit=&patents_us=on

[89] US patent result set: "vaccine temperature monitoring time temperature indicator," FreePatentsOnline. https://www.freepatentsonline.com/result.html?query_txt=vaccine+temperature+monitoring+time+temperature+indicator&submit=&patents_us=on

[90] "Pharmacy Temperature Monitoring," Envigilance. https://envigilance.com/blog/pharmacy-temperature-monitoring/

[91] "How Do You Detect a Cold-Chain Excursion Quickly?," SenseAnywhere. https://www.senseanywhere.com/how-do-you-detect-a-cold-chain-excursion-quickly/

[92] "The Tip of the Iceberg" — economic analysis of cold-chain losses, IQVIA EMEA whitepaper. https://www.iqvia.com/-/media/iqvia/pdfs/emea/library/whitepaper/tip-of-the-iceberg-economic-and-environmental-impact-of-the-vaccine-cold-chain.pdf

[93] Matthias et al., "Freezing temperatures in the vaccine cold chain: a systematic literature review," Vaccine (2007) — PubMed record. https://pubmed.ncbi.nlm.nih.gov/17382434/

[95] Matthias et al., Vaccine (2007) — ScienceDirect record. https://www.sciencedirect.com/science/article/abs/pii/S0264410X07002289

[96] "VFC Data Loggers & Vaccine Temperature Monitoring," Plumsense. https://www.plumsense.com/vfc-data-loggers-vaccine-temperature-monitoring/

[97] Follow-up review on freezing exposure in the vaccine cold chain, Vaccine (2017/18) — ScienceDirect. https://www.sciencedirect.com/science/article/pii/S0264410X16309471

[98] "Vaccine Temperature Monitoring Best Practices," Dickson Data. https://dicksondata.com/vaccine-temperature-monitoring-best-practices?srsltid=AfmBOoo5iuJKeEqiMqB4VIP9Zkk3sABM6BLpob_JV2TWzNSwEDP2rVkl

[100] Immunization Handbook, WHO (India SEARO). https://cdn.who.int/media/docs/default-source/searo/india/publications/immunization-handbook-107-198-part2.pdf

[102] SpotSee corporate site — ShockWatch/ShockLog/SpotBot condition indicators. https://spotsee.io/?srsltid=AfmBOorJnZYBWuIn2rmJc5yOdeh4_79WffstH3on5b7I0gU1-Er1Z0B9

[103] SpotSee manufacturer page, Mouser Electronics. https://www.mouser.hk/manufacturer/spotsee/?srsltid=AfmBOopVSpaCGDzn17dWrvKyheTT8pasg9tju7X6u-dp5_lxDBRYq7D4

[104] SpotSee supplier center, Digi-Key Electronics. https://www.digikey.com/en/supplier-centers/spotsee

[107] Orbit Reader 20 review, AFB AccessWorld (afb.org/aw/19/1/15167). https://afb.org/aw/19/1/15167

[108] "Orbit Reader 20: The Most Inexpensive Braille Display," National Federation of the Blind. https://nfb.org/orbit-reader-20-most-inexpensive-braille-display

[109] Orbit Reader / Transforming Braille Group page, RNIB. https://www.rnib.org.uk/living-with-sight-loss/assistive-aids-and-technology/reading-and-writing/introducing-the-orbit-reader-20/

[110] "Braille Me and Orbit Reader Braille Display Comparison," Perkins School for the Blind. https://www.perkins.org/resource/braille-me-and-orbit-reader-braille-display-comparison/

[111] Orbit Reader 20 product page, American Printing House (tech.aph.org/or20). https://tech.aph.org/or20/

[112] Orbit Reader 20 Plus listing, A.T. Guys. https://atguys.com/products/orbit-reader-20-plus-braille-display-book-reader-and-note-taker?srsltid=AfmBOoonI2XdtKbAGe3N9bE8whj9mZlfYlMjKKt9ZY662qrxB4VDOvdZ

[113] Orbit Reader 20 listing (manufacturer description), MaxiAids. https://www.maxiaids.com/product/orbit-reader-20

[114] US 7,884,620 — Sensor for measuring moisture and salinity (full text, FreePatentsOnline). https://www.freepatentsonline.com/7884620.html

[115] US 11,598,743 — Soil monitoring sensor including single probe and temperature compensation (full text). https://www.freepatentsonline.com/11598743.html

[116] US 7,102,526 — Electronic time-temperature indicator and logger (full text). https://www.freepatentsonline.com/7102526.html

[117] US 10,887,735 — Vaccine monitoring system (full text). https://www.freepatentsonline.com/10887735.html

[118] US 7,421,349 — Bearing fault signature detection (full text). https://www.freepatentsonline.com/7421349.html

[119] US 10,740,084 — SoC-assisted resilient boot (full text). https://www.freepatentsonline.com/10740084.html

[120] US 9,953,166 — Method for securely booting a target processor using a secure root of trust (full text).
https://www.freepatentsonline.com/9953166.html

[121] US 11,042,609 — Systems and methods for secure element registration and provisioning (full text).
https://www.freepatentsonline.com/11042609.html